



Export controls and innovation performance: Unravelling the complex relationship between blacklisted Chinese firms and U.S. suppliers

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Abstract

Export controls are crucial for protecting domestic economic interests globally. However, there is a lack of consensus regarding their impact on innovation performance. This study contributes to the literature by examining the effects of the U.S. entity list on the innovation performance of blacklisted Chinese firms and related industries, as well as the reverse shock on U.S. suppliers. Using propensity score matching and a difference-in-differences approach, we found that export controls stimulate technological innovation in blacklisted firms and generate industry-wide spillover effects in the sanctioned country. Particularly, firms heavily reliant on imports from the United States and those solely dependent on China's domestic market experience significant effects. However, export controls have a negative impact on the innovation performance of U.S. suppliers. We examine the mechanisms driving these effects, including government grants, R&D investment and firm performance. Our work offers valuable insights into the complex relationship between export controls and innovation performance, highlighting the differentiated impacts on blacklisted firms, related industries and U.S. suppliers.

KEYWORDS

export controls, firm innovation, spillover effects, U.S. entity list

1 | INTRODUCTION

Export control is a common means for governments to use laws and administrative measures to manage and control exports of specified products or technologies to other countries (Bown, 2019; Meijer, 2013, 2015). In the past few decades, due to concerns related to national security and diplomatic interests, the United States has implemented a series of export control policies, especially against China – whose economic and military strength is at a stage of rapid development – to prevent potential or explicitly sensitive technology from flowing to China. This has amplified the United States's geo-economic strategy to maintain its own technological leadership. The entity list system is an important part of the U.S. export administration's regulations and serves the country's strategic interests and foreign policy. It is controlled by the Bureau of Industry and Security in the U.S. Department of Commerce and used to list entities that the U.S. government believes to be involved in or have a significant risk of participating in activities that harm U.S. national security or economic interests (Bu, 2020). This means that blacklisted firms need to apply for special permission to import goods or technologies from the United States. Simultaneously, U.S. exporters also need to complete a strict and complex export licensing process when exporting, re-exporting or transferring items to blacklisted firms.

In total, 540 entities were added to the entity list between 1997 and 2021, and as of the end of 2021, they are still subject to export control sanctions. Notably, the number of blacklisted Chinese entities has significantly risen in recent years, particularly from 2018 onwards (refer to Table 1). Against the backdrop of the United States's escalating export controls on China, imitation and innovation paths by importing products and technology for sanctioned Chinese firms are restricted. This restriction has a substantial impact on the innovation of firms on the entity list, as well as on the corresponding Chinese and U.S. firms, through the transmission of domestic and international production networks. In broader terms, the expanding export controls imposed by the United States on China have profound implications for innovation pathways and the functioning of production networks. This impact extends to firms listed on the entity list, as well as those associated with Chinese and U.S. entities.

Recent literature on export control policies has primarily focused on investment (Li et al., 2023; Seyoum, 2017), trade effects (Richardson & Sundaram, 2013), consumer welfare (Yang et al., 2004) and the competitiveness of countries and industries (Chiang, 1987; Mineiro, 2011).

TABLE 1 Number of blacklisted Chinese entities from 1997 to 2021.

Year	Number of sanctioned entities	Year	Number of sanctioned entities	Year	Number of sanctioned entities
1997	1	2006	0	2015	21
1998	0	2007	0	2016	18
1999	6	2008	4	2017	1
2000	0	2009	4	2018	54
2001	14	2010	19	2019	107
2002	0	2011	14	2020	149
2003	0	2012	4	2021	81
2004	0	2013	13		
2005	0	2014	30		

Source: Authors' own calculation based on the U.S. Bureau of Industry and Security.



However, studies on the effect of export control policy on firm-level innovation activities are relatively sparse. In this study, we identify the Chinese public firms that have been blacklisted by the U.S. entity list between 1997 (the year the entity list was implemented) and 2021, using propensity score matching (PSM) techniques and a multiple-period differences-in-differences (DID) approach. We aim to examine the policy's impact on innovation in Chinese public firms that have been blacklisted by the entity list and other Chinese public firms in the same industries and document the reverse impact of export control on U.S. suppliers of sanctioned Chinese public firms.

Empirical analysis of the data in the 1997–2021 period shows that the U.S. entity list has a significant positive influence on the innovation performance of blacklisted Chinese firms, especially for firms whose production activities and operations depend on imports from the United States. This induces the allocation of Chinese government grants, increases blacklisted firms' R&D investment and makes the sanctioned firms shift their development focus to their main business. The implementation of the U.S. entity list promotes the innovation performance of Chinese publicly listed firms in the same industries and has a heterogeneous effect on whether firms rely solely on China's domestic market. Furthermore, U.S. export control has a reverse impact on U.S. suppliers by decreasing income, profit and return on capital employed.

This paper makes the following important contributions to the literature. First, we add to the developing literature on the consequences of international trade barriers by examining the impact of export control on blacklisted Chinese firms' innovation performance. The impact of export control barriers at the firm level is rarely explored. Thus, we examine related industry diffusion effects and the reverse impact of the entity list on U.S. suppliers to build a comprehensive analysis regarding the policy's effect. Second, the results presented in this paper extend the literature on the determinants of firm innovation. Related literature tends to focus on the effect of external policies, such as economic policies and government subsidies (Bronzini & Piselli, 2016; Chava et al., 2013; Shao & Xiao, 2019), or firms' internal factors, such as managerial background characteristics and financial constraints (Chen et al., 2019; Gorodnichenko & Schnitzer, 2013; Hambrick & Mason, 1984). Few studies have provided direct evidence on how export control policies affect firm innovation.

The rest of the paper is organised as follows. Section 2 presents a review of the relevant literature. In Section 3, we describe our data and empirical strategy. Sections 4 and 5 present our empirical findings on the impact of the entity list on blacklisted Chinese firms and U.S. suppliers. Section 6 concludes the paper.

2 | REVIEW OF RELATED LITERATURE

Historically, as discussed below, research has predominantly focused on examining the impact of export controls on innovation from the perspectives of policy-implementing countries and sanctioned countries. However, several studies reviewed in this section reveal that there is no consensus regarding the effect of export control on innovation performance.

For policy-implementing countries, export control is often seen as a means to protect national security and maintain technological advantages (Stinnett et al., 2011). However, this type of economic sanction is likely to fail in achieving its objectives (Pape, 1997, 1998) and can introduce excessive costs to the implementing country (Bown, 2020). This can lead to a loss of market share and enhanced competitive advantage for international competitors (Mineiro, 2011), while the implementing country can experience a decline in domestic jobs and damage to domestic welfare (Yang et al., 2004). The stability of the supply chain may also be questioned, resulting in additional costs for domestic industry development (Bown, 2020).

Consequently, the international competitiveness of domestic firms in related industries can decline, hindering their participation in global industrial supply chains and impeding further innovation (Parkhe, 1992).

Existing studies indicate that sanctioned countries' innovation performance is negatively affected by export control policies and restricted import procedures (Hosoe, 2021; Marukawa, 2013). As imports play a crucial role in promoting firms' innovation performance, export control policies that ban products containing frontier technologies can limit knowledge spillovers and follow-on innovation channels in sanctioned countries (Chiang, 1987).

Export control can also increase innovation costs and impede the upgrading of industrial chains, particularly for developing countries that rely on international outsourcing and entrusted manufacturing within global value chains. Upgrading high-end links, such as design, development and marketing, becomes challenging (Hummels et al., 2001; Timmer et al., 2014). Additionally, export control destabilises the sanctioned countries' supply chain network, introduces uncertainty for local and foreign-invested firms, reduces overall productivity, increases production costs, decreases profits and limits firms' research and development capacity (Hosoe, 2021). Furthermore, export control can interfere with trade and technical cooperation between sanctioned countries and countries aligned with the policy-implementing country, thereby impacting the competitiveness, upgrading and R&D internationalisation of the sanctioned country's corresponding industries (Choi & Niculescu, 2006).

However, some studies suggest that export control may benefit innovation in sanctioned countries. First, appropriate export control can reduce the controlled country's dependence on foreign technology and encourage innovative activities to improve production efficiency (Porter, 1996). Export control also limits product variety in sanctioned markets, reducing product market competition intensity and providing domestic firms with more room for survival and development. This counterbalances the negative impact on innovation caused by decreased profit margins (Dorn et al., 2020; Liu et al., 2021). The increasing instability of international supply chains caused by export control prompts more firms in sanctioned countries to relocate their supply chains domestically, creating additional market demand and stimulating innovation and development in related industries (Piva & Vivarelli, 2007). Moreover, export control increases market competition pressure on directly impacted firms, motivating them to invest in R&D and engage in innovation activities to improve product competitiveness, maintain market position, capture a larger market share and increase profits (Xu et al., 2022).

However, most studies on the impact of export controls on innovation performance have focused on national and industrial levels, relying on qualitative analyses. There is a lack of research at the micro level that comprehensively explores the mechanisms and impacts of export control using a mature analysis framework and quantitative research methodology. Our empirical findings indicate that the U.S. entity list has a stronger effect in promoting innovation performance in blacklisted Chinese firms compared to non-blacklisted firms. Furthermore, we identify and discuss the impact of export control on the industrial supply chain, finding that innovation promotion has spread to relevant Chinese industries, while the entity list has restrained U.S. suppliers' innovation performance due to a reverse effect.

3 | DATA AND EMPIRICAL STRATEGY

3.1 | Data sources

The data for our study were gathered from diverse sources. First, we collected information on invention patents held by Chinese public firms from the China National Intellectual Property



Administration (CNIPA),¹ a Chinese government agency responsible for intellectual property rights examination, registration and adjudication. CNIPA conducts trademark registration and patent examinations, offering a patent search function covering all publicly available patent applications within China. This functionality allows the retrieval of patent information registered by any company in the country. The dataset encompassed patent names, application dates, publication data, abstracts and international patent classifications (IPC). We utilised Chinese public firms' names to search the database, compiling the number of invention patents and IPC varieties by year.

Second, we obtained the '*Entitylist*' dummy variable for the Chinese component from the U.S. Bureau of Industry and Security. To align Chinese firms' English names in the entity list, we translated them into Chinese names and cross-referenced them with the Chinese public firms' database. This process identified blacklisted Chinese public firms. Despite the complexity of removing entities from the list, we addressed this by retrieving historical files from the U.S. Bureau of Industry and Security. Our analysis showed that, as of the end of 2021, 31 Chinese entities had been removed, with only two being publicly traded Chinese firms – ZTE (removed in 2017) and Jerry Oil (removed in 2019). We treated these entities differently in data processing, setting the *EntityList* variable to 0 after 2017 and 2019 respectively. This treatment was also applied to their associated individual U.S. suppliers. We additionally gathered information on 50 sanctioned Chinese public firms with at least 1 year of public listing.

Furthermore, we accessed data on Sino–U.S. supply relationships from the FactSet Supply Chain Relationships database. This information helped identify U.S. suppliers of Chinese firms on the entity list, leading to the creation of the '*US_EntityList*' dummy variable for the U.S. component of our analysis. The FactSet database, recording supply relationships since 2003, includes variables such as company names, ISIN codes, supply relationship start and end dates and relationship descriptions. We manually searched the database based on the English names of sanctioned Chinese firms, avoiding potential mismatches due to case or punctuation differences. ISIN codes were used to determine each firm's country of origin, minimising the risk of duplicate firm names from different countries. Based on the timing of sanctions on Chinese firms and supply relationship dates, we identified U.S. suppliers that established relationships with these Chinese firms before sanctions. These suppliers, impacted by export control policies, were unable to continue supplying products to their Chinese customers. Finally, data for control variables were obtained from the China Stock Market and Accounting Research (CSMAR) database and Wharton Research Data Services (WRDS), providing financial data from annual reports of Chinese and U.S. public firms respectively. This diverse data collection resulted in a comprehensive dataset for our empirical analysis.

3.2 | Data description

Table 2 presents the descriptive statistics for the variables used in this study. The dependent variables used to measure firms' innovation performance were the number of invention patents and the variety of patents' IPC classifications. The International Patent Classification (IPC) serves as a globally recognised tool for categorising patents based on various technological fields. Past research indicates that a higher number of IPC classifications associated with a patent implies a broader coverage across fields, signifying greater patent value and quality (Hvide & Jones, 2018;

¹<https://www.cnipa.gov.cn/>.

TABLE 2 Descriptive statistics.

Variable	Obs	Mean	Std.dev.	Min	Max
Chinese public firms directly affected by the entity list					
<i>lnPatent</i>	1532	2.4454	2.0632	0.0000	9.0873
<i>lnIPC</i>	1532	3.0480	2.3697	0.0000	9.9215
<i>Entitylist</i>	1532	0.0633	0.2436	0.0000	1.0000
<i>lnAsset</i>	1527	22.3735	1.5136	18.7354	27.9609
<i>lnAge</i>	1532	2.5612	0.5979	0.0000	3.5835
<i>TFP</i>	1445	1.0598	0.5374	-2.6479	3.1499
<i>FC</i>	1527	4.5770	1.7498	0.8057	12.4108
<i>lnCapital</i>	1447	14.4328	1.0524	11.3205	18.6536
<i>lnSalecost</i>	1523	21.6060	1.7023	16.4248	27.2029
<i>lnRD</i>	902	18.7650	1.7718	11.8927	25.0252
Chinese firms in related industries affected by the entity list					
<i>lnPatent</i>	24,901	1.8294	1.6580	0.0000	9.0984
<i>lnIPC</i>	24,901	2.4133	2.0125	0.0000	10.1867
<i>Entitylist</i>	24,901	0.1239	0.3294	0.0000	1.0000
<i>lnAsset</i>	24,872	21.5773	1.2685	13.0760	28.5016
<i>lnAge</i>	24,900	2.6381	0.5543	0.0000	3.8286
<i>TFP</i>	23,395	0.4493	0.5947	-11.5344	3.8193
<i>FC</i>	24,872	3.5942	1.4081	-3.1248	13.3650
<i>lnCapital</i>	23,425	14.1871	0.8634	9.7193	19.7052
<i>lnSalecost</i>	24,806	20.8513	1.4508	13.9585	28.2097
<i>lnRD</i>	17,640	17.9752	1.4458	7.0901	24.4103
American suppliers of Chinese public firms affected by the entity list					
<i>lnPatent</i>	19,174	0.4050	1.0330	0.0000	8.0488
<i>lnIPC</i>	19,174	0.4943	1.2074	0.0000	8.6486
<i>Entitylist</i>	19,174	0.0150	0.1214	0.0000	1.0000
<i>lnAsset</i>	18,395	5.4313	3.0238	-6.9078	13.5295
<i>TFP</i>	15,491	2.7334	0.8180	-6.0733	8.2196
<i>FC</i>	15,460	-0.0055	0.3605	-28.4825	20.1475
<i>lnCapital</i>	16,066	6.0611	1.4587	-2.3026	15.1881
<i>lnSalecost</i>	17,396	4.6622	2.7895	-6.9078	12.2233
<i>lnRD</i>	10,890	2.8327	2.3957	-6.9078	10.3597

Source: Authors' own calculation based on Google Patent, CSMAR and WRDS.

Lerner, 1994). This approach allows for a comprehensive evaluation of firms' innovation capabilities and the scope of their innovation activities.

At the firm level, the control variables included the natural logarithm of total assets (*lnAsset*) and the natural logarithm of the firm's age (*lnAge*). These variables capture the significance of accumulated experience and resource acquisition in shaping firms' innovation capabilities (Audretsch & Acs, 1991; Balasubramanian & Lee, 2008). Additionally, we included total factor



productivity (TFP) as a control variable to account for heterogeneity in productivity, calculated using the method proposed by Akerberg et al. (2015). Specifically, we assumed that the output of the firm was determined by a Cobb–Douglas production function as follows:

$$Y_{it} = TFP_{it} L_{it}^{\alpha} K_{it}^{\beta} \quad (1)$$

where subscript i represents the firm, t represents the year, Y represents output and L and K represent the firm's labour and capital inputs respectively.

The firm-level TFP was calculated using the logarithmic transformation of Equation (1) as follows:

$$\log(TFP_{it}) = \log(Y_{it}) - \alpha \log(L_{it}) - \beta \log(K_{it}) \quad (2)$$

Following Cozza et al.'s (2015), we used the sales revenue of the firms as a proxy for output, the number of employees as a proxy for labour input and total assets as a proxy for capital. In line with the LP-based ACF correction method for TFP calculation, we used raw material input data as a proxy variable for input (Fons-Rosen et al., 2021; Levinsohn & Petrin, 2003). The data were obtained from the CSMAR database and the Compustat database at WRDS.

To measure firms' financing constraints, we included the variable financing constraints (FC). Due to data limitations, we calculated the financing constraints for Chinese public firms using the method outlined by Hadlock and Pierce (2010), and for American public firms using the method proposed by Whited and Wu (2006).

Furthermore, we considered firms' sales costs ($\ln Salecost$) as a measure of operational expenses, which is publicly disclosed data for public firms. Finally, we included firms' capital intensity ($\ln Capital$) in the regression model, as it is an important factor that promotes firm innovation (Acs & Audretsch, 1987). These control variables were included to account for various factors that could potentially influence firms' innovation performance.

Figure 1 illustrates the trends in the average innovation performance of Chinese public firms that have been blacklisted, compared to firms in the same industry. For each year, we calculated the average number of patent applications and the variety of IPC classifications for blacklisted public firms and firms in the same industry. In Figure 1, the lines represent the average innovation performance of blacklisted public firms, while the columns represent the average innovation performance of industries affected by the entity list. It is evident that the innovation performance of blacklisted public firms is consistently higher than the industry average for most years. This finding suggests that Chinese public firms that have been blacklisted by the entity list play a leading role in their respective industries. These firms exhibit strong innovation potential, promising future development prospects and exceptional international competitiveness. Although there was a decline in their innovation performance following the financial crisis in 2008, the overall leading position of these blacklisted firms has remained unchanged. This indicates their enduring strength and continued influence within their respective industries.

Figure 2 presents the average innovation performance of U.S. suppliers whose Chinese customers had been blacklisted by the entity list. The graph shows the average number of invention patents and variety of IPCs over time, with both measures declining after 2018. Importantly, it is worth noting that the strength of U.S. export control against Chinese products also significantly intensified during the same period. This observation prompted us to explore whether there is a causal relationship between the changing trend in innovation performance and the strengthening

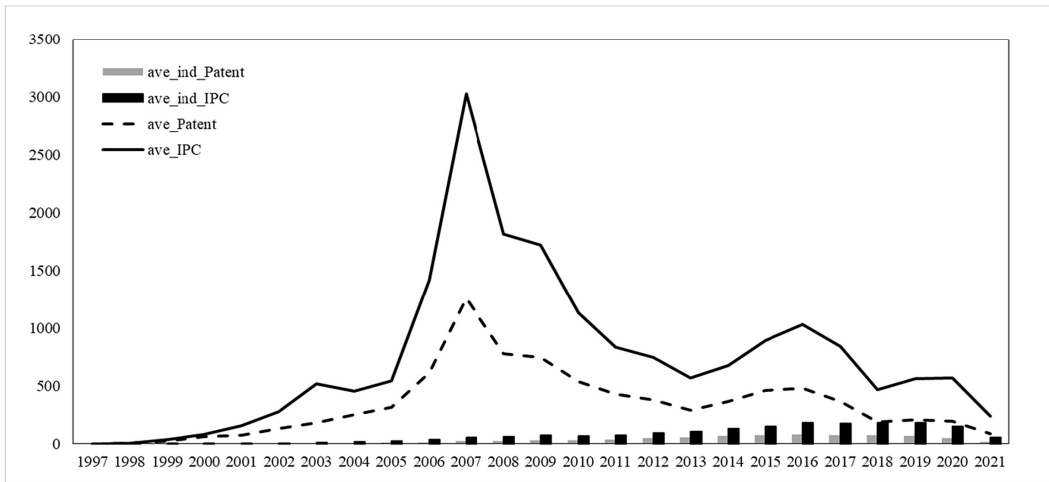


FIGURE 1 Innovation performance of firms impacted by the U.S. entity list from 1997 to 2021.

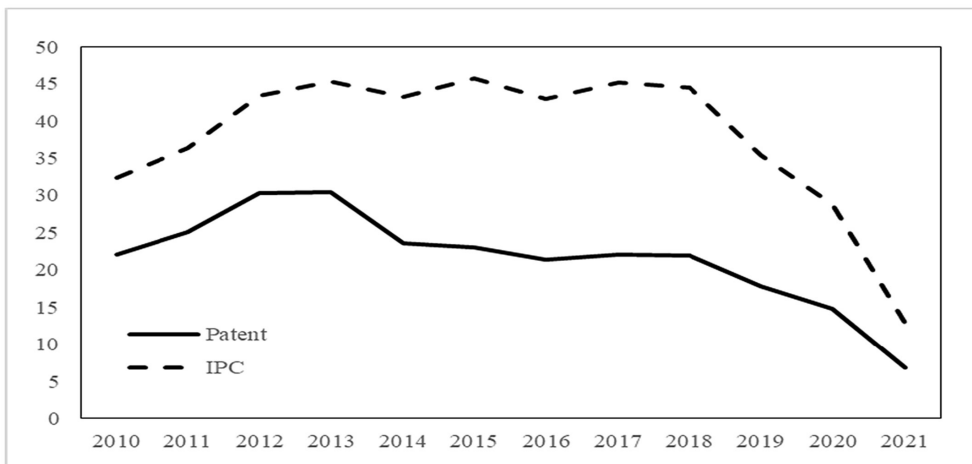


FIGURE 2 Innovation performance of the U.S. suppliers from 2010 to 2021.

trend in export control. By examining the concurrent changes in these trends, we were able to gain insights into the potential causal relationship between these factors.

We directed our attention to industries targeted by U.S. export control, and Table 3 delineates the industry distribution of sanctioned Chinese firms and their corresponding U.S. suppliers. China's industry classification adheres to the National Economic Industry Classification, whereas that of the U.S. follows the Global Industry Classification Standard.

The percentage of Chinese sanctioned firms and their affected U.S. suppliers within each industry was calculated relative to the total number of Chinese sanctioned firms and impacted U.S. suppliers respectively. Notably, both affected entities in China and the United States were concentrated in knowledge-intensive industries. The computer, communication and other electronic equipment manufacturing industry exhibited the highest distribution of sanctioned Chinese firms (48.57%), while the semiconductor industry registered the highest percentage of affected



TABLE 3 Industry distribution of the top 10 Chinese and U.S. firms affected by the entity list.

China			US		
Industry name	Industry code	Proportion	Industry name	Industry code	Proportion
Computer, communication and other electronic equipment manufacturing	C39	48.5714	Semiconductors	45301020	16.9811
Software and information technology service industry	I65	14.2857	Application Software	45103010	8.8050
Electrical machinery and equipment manufacturing industry	C38	8.5714	Semiconductor Equipment	45301010	6.2893
Automobile manufacturing industry	C36	5.7143	Electronic Components	45203015	5.0314
Chemical raw materials and chemical products manufacturing industry	C26	5.7143	Industrial Machinery	20106020	4.4025
Textile industry	C17	2.8571	Systems Software	45103020	3.7736
Special equipment manufacturing industry	C35	2.8571	Communications Equipment	45201020	3.7736
Civil engineering construction industry	E48	2.8571	Biotechnology	35201010	3.1447
Pharmaceutical manufacturing industry	C27	2.8571	Pharmaceuticals	35202010	3.1447
Internet and related services industry	I64	2.8571	Aerospace & Defence	20101010	2.5157

Source: Authors' calculation based on data from the U.S. Bureau of Industry and Security and the Fact set.

U.S. suppliers (16.98%). Given the pivotal role of innovation in these industries' development, a more in-depth investigation is warranted to assess the impact of export control on the innovation performance of both the sanctioned country (China) and the policy-implementing country (United States).

3.3 | Empirical strategy

3.3.1 | PSM technique

To examine the impact of export control on the innovation performance of Chinese blacklisted public firms, we employed propensity score matching and a multiple-period difference-in-differences approach.

Using the PSM technique, we identified corresponding firms for each blacklisted firm that were unaffected by the entity list to ensure the accuracy of the comparison. To ensure the robustness of the matching, we analysed released information from the U.S. Bureau of Industry and Security outlining the reasons for firms being sanctioned by the United States. From our analysis, we identified several factors that contributed to firms being sanctioned. These factors include

political behaviour associated with the Chinese government that violates the national security and diplomatic interests of the United States; firms' international competitiveness posing a threat to the economic security and interests of the United States; and firms exhibiting irrationality in their international business activities.

Thus, our matching variables focused on firms' international competitiveness, their overseas business activities, their connection with the Chinese government and their industry. By considering these factors, which are explained in detail below, we aimed to ensure that the matched firms were as similar as possible to the blacklisted firms, allowing for a reliable comparison of their innovation performance.

International competitiveness of the firms

The international competitiveness and innovative potential of firms are crucial considerations for the United States when selecting targets for export controls. Prominent Chinese public firms such as Huawei, ZTE, iFlytek, 360 and others have been sanctioned over the years. These companies are recognised leaders in their respective industries and have played a significant role in breaking technological monopolies, bolstering China's national security and enhancing the country's global competitiveness through their advanced technologies and robust innovation capabilities.

Furthermore, the international competitive advantages of these firms are key factors that attract foreign collaborations, establish supply relationships and create business dependencies. For the United States, which seeks to maintain its position as a technological leader and aims to mitigate risks associated with China, these companies present challenges to U.S. intervention policies. Consequently, the international competitiveness of Chinese firms emerges as a crucial determinant in whether they become subject to U.S. export controls.

To assess firms' international competitiveness, we adopted variables that impact business performance, as studied by Brandt et al. (2017) and Beerli et al. (2021). We selected 'lnCapital' and 'lnAsset' to measure a firm's development position as capital intensive or labour intensive and to determine its size. Capital-intensive firms tend to focus more on R&D, which influences their international competitiveness (Audretsch & Acs, 1991). Additionally, firm size is associated with market share and strength (Acs & Audretsch, 1987). We also considered 'Sale_profit' and 'ROE' as variables to gauge a firm's profitability, which are important indicators of competitiveness. Furthermore, we included TFP as a matching variable, reflecting the firm's technological level.

Overseas operations of the firms

Our investigation revealed that a critical determinant influencing the susceptibility of firms to U.S. export controls is their expansion into international markets. Transactions involving U.S. items by Chinese firms with entities or nations that oppose U.S. diplomatic interests or pose threats to national security constitute significant triggers for sanctions. Simultaneously, the imposition of export controls represents a strategic measure employed by the United States to impede China's progress and widen the technological disparity between the two nations. Chinese public firms engaged in overseas operations gain advantages in assimilating advanced technology and participating in cross-border knowledge and talent exchanges. While contributing significantly to China's independent innovation development, these firms inevitably emerge as formidable obstacles to the effective implementation of U.S. sanction policies. Moreover, involvement in overseas business and participation in international competition reflect a firm's robust international competitiveness. Consequently, a firm's export behaviour and other forms of overseas business operations play a pivotal role in determining its susceptibility to U.S. sanctions.



Due to confidentiality constraints on data, the China Customs Import and Export Database, documenting the import and export activities of Chinese enterprises, only encompasses publicly disclosed data from 2000 to 2016. However, the majority of U.S. export control sanctions on China were imposed after 2018. Subsequent to 2016, information on the overseas revenue of Chinese public firms is available solely through their annual reports. Unfortunately, these data lack differentiation between countries or regions, and its disclosure is not mandatory. Given these limitations, our most diligent effort to assess the overseas business development of Chinese public firms involved amalgamating these two databases and creating a variable, denoted as '*Overseas*', indicating whether the firm engaged in overseas business. Specifically, if a firm exhibited export behaviour before 2016 or actively disclosed overseas revenue data, this variable was set to 1; otherwise, it was set to 0.

Firms' connection with the Chinese government

Owing to political disparities, alignment with the policies of the Chinese government is perceived as a threat to the national security and diplomatic interests of the United States, constituting a significant impetus for the imposition of sanctions on Chinese firms by the United States. In this study, we utilised the degree of affinity between Chinese firms and the Chinese government as a variable to gauge the probability of these firms endorsing the actions of the Chinese government. Notably, in China, the Communist Party of China (CPC) holds the highest echelons of political leadership and serves as the representative of the Chinese government. To assess the proximity of the relationship between a firm and the Chinese government, we amassed data on the count of party members occupying senior leadership positions within the firm, as reported in its corporate documentation. This variable, denoted as '*InParty*', functioned as a matching variable. A higher count of party members in the senior leadership of a firm signifies a more intimate connection between the firm and the Chinese government, thereby escalating the likelihood of the firm endorsing governmental actions (Li et al., 2008; Li & Chan, 2016) and subsequently becoming a target for U.S. sanctions.

Industry-specific characteristics

Table 3 highlights the industry-specific nature of U.S. export control sanctions against China, with many sanctioned public firms belonging to high-tech industries. To ensure the accuracy of our matching, we restricted the sample range to firms in the same industry as the sanctioned firms.

To estimate the propensity score, we performed a logit model based on the year before a firm was blacklisted. Following the approach of Huang and Zhang (2017), we employed a one-to-one nearest-neighbour matching strategy to identify corresponding firms for the control group. The estimation model is as follows:

$$\Pr(\text{ExportControl}_{it} = 1) = \phi(\beta_1 \ln \text{Asset}_{t-1} + \beta_2 \ln \text{Capital}_{t-1} + \beta_3 \text{TFP}_{t-1} + \beta_4 \text{Saleprofit}_{t-1} + \beta_5 \text{Roe}_{t-1} + \beta_6 \text{Overseas}_{t-1} + \beta_7 \ln \text{Party}_{t-1}) \quad (3)$$

where *ExportControl* is a dummy variable that takes the value of 1 if the firm was affected by U.S. export controls; 0 otherwise.

The propensity score serves as a predictive measure for the probability of Chinese public firms encountering U.S. export control sanctions in year *t*. The effectiveness of our matching approach is evidenced in Table 4, which presents the results of the balance check. Utilising a one-to-one nearest-neighbour matching technique, we successfully paired each of the 50 Chinese public firms affected by U.S. export control sanctions with a control firm possessing the most analogous

TABLE 4 Propensity score matching (PSM) balance check.

Variable	Unmatched	Mean		%bias	t-test	
	Matched	Treat	Control		t	p > t
lnAsset	U	23.14	22.089	73.6	5.51	.000***
	M	23.14	23.075	4.6	0.22	.827
lnCapital	U	14.836	14.589	27.6	1.81	.071*
	M	14.836	14.88	-4.9	-0.22	.824
TFP	U	0.9554	1.0774	-21.7	-1.49	.137
	M	0.9554	0.8917	11.3	0.61	.545
Sale_profit	U	1.4e+09	5.6e+08	21.9	1.80	.073*
	M	1.4e+09	1.1e+09	8.4	0.47	.639
Roe	U	-0.0724	0.0161	-3.2	-0.16	.870
	M	-0.0724	0.07	-5.2	-1.55	.125
Overseas	U	0.92	0.9305	-4.0	-0.29	.771
	M	0.92	0.9	7.6	0.35	.730
lnParty	U	0.8398	0.6173	27.4	2.01	.044**
	M	0.8398	0.7498	11.1	0.53	.601

Note: Standard errors are in parentheses. *** $p < .01$; ** $p < .05$ and * $p < .1$.

characteristics. Following the matching process, the ratio of firms in the treatment and control groups became 1:1.

As shown in Table 4, the differences between the treatment and control groups in most matching variables significantly decreased after matching. Overall, the standardised bias across covariates after matching was within ± 10 . Therefore, after matching, these groups showed similar characteristics that could be affected by the entity list 1 year before its implementation, suggesting that our matching is reasonable.

3.3.2 | Multiple-period DID specification

To investigate the impact of export controls on blacklisted Chinese firms, we conducted a multiple-period DID analysis using the following specification.

$$Y_{ft} = \beta_1 + \beta_2 \text{Entitylist}_{ft} + X_{ft}\gamma + \theta_f + \mu_{it} + \varepsilon_{fit} \quad (4)$$

where the subscripts f , i and t denote the firm, industry and year respectively. Y_{ft} represents the dependent variables, which encompasses $\ln patent$ and $\ln ipc$, for firm f at time t . Entitylist_{ft} is a dummy variable which takes a value of 1 if firm f is affected by the entity list in year t , 0 otherwise. X_{ft} represents a set of time-varying firm-level control variables known to potentially influence the firm's innovation performance, as outlined in Section 3.2. θ_f represents the firm fixed effects, μ_{it} denotes industry-year fixed effects and ε_{fit} is the usual error term.

The coefficient β_2 , in Equation (4), captures the average differences in innovation performance between firms affected by the entity list and those unaffected by it.



4 | EMPIRICAL ANALYSIS AND FINDINGS

First, we explored the impact of the U.S. entity list on blacklisted Chinese public firms' innovation performance.

4.1 | Baseline specification

A crucial assumption for the DID estimation to be valid is that the treatment and control groups follow parallel trends before a firm is blacklisted. To identify the validity of the multiple-period DID, we began by estimating an event study for the parallel trend test:

$$Y_{ft} = \sum_{k=-4, k \neq -1}^4 D_{ft}^k \delta_k + X_{ft} \gamma + \theta_f + \mu_{it} + \varepsilon_{fit} \quad (5)$$

where the Y_{ft} is the dependent variables, including $\ln Patent$ and $\ln IPC$ in firm f at time t .

The dummy variable D_{ft}^k signifies the occurrence of the firm being blacklisted by the U.S. entity list. Specifically, $D_{ft}^{-4-} = 1$ indicates the years before and in the fourth year preceding the firm's inclusion in the entity list, while $D_{ft}^{-4-} = 0$ pertains to other years. The dummy for $k = -1$ was omitted to ensure that the post-treatment effects are referenced relative to the period preceding the addition of Chinese firms to the entity list, aligning with the methodology employed by Berniell et al. (2021) and Chen and Lan (2020). Our main coefficient of interest was δ_k , which identifies the effect of export control k years following its implementation. X_{ft} represents a set of time-varying firm-level control variables that might affect the firm's innovation performance. θ_f is the firm fixed effect, which captures all the time-invariant firm characteristics that might influence the innovation performance of firms. Additionally, industry-year fixed effects, μ_{it} , were included to control time- and industry-specific characteristics that might affect innovation performance.

Figure 3 showcases the estimates derived from Equation (2) to elucidate the dynamic impact of export control on innovation performance. The graphical representation indicates an absence of a discernible increase or decrease in the trend of innovation performance leading up to the blacklisting of firms by the U.S. entity list. The coefficient value converges towards 0, signifying a lack of statistically significant differentiation in innovation performance between the treatment group (blacklisted firms) and the control group (non-blacklisted firms) prior to the implementation of export control measures.

However, after the firms were subjected to export control, a significant difference in innovation performance emerged between the sanctioned and non-sanctioned firms. This observation satisfies the parallel trend assumption, indicating that the groups were comparable in terms of their innovation performance trends before the implementation of export control measures. Notably, in the second year following the imposition of U.S. export control sanctions, a significant disparity in innovation performance between the sanctioned and non-sanctioned firms becomes evident. This finding highlights the impact of export control on the innovation performance of the sanctioned firms.

We employed the multiple-period DID approach as the benchmark regression method to investigate the impact of export controls on Chinese firms' innovation performance. We compared the differences in innovation performance between blacklisted and non-blacklisted Chinese firms before and after the implementation of export control measures. The results of the benchmark regression are presented in Table 5.

The coefficients of the variable 'Entitylist' in all specifications were positive and statistically significant. This finding suggests that blacklisted Chinese firms experience notable growth in

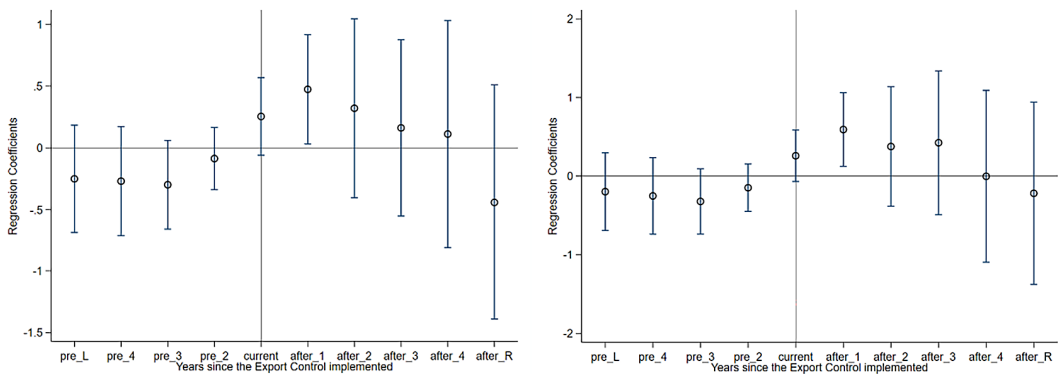


FIGURE 3 Parallel trend test of Chinese public firms blacklisted by the entity list. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/twe.13570)]

TABLE 5 Effect of the entity list on the innovation performance of blacklisted Chinese firms.

	(1)	(2)
	<i>lnPatent</i>	<i>lnIPC</i>
<i>EntityList</i>	0.4266** (0.1633)	0.4711*** (0.1688)
<i>lnAsset</i>	1.7525 (1.1265)	1.4088 (1.2510)
<i>lnAge</i>	0.0775 (0.4444)	−0.0962 (0.5284)
<i>TFP</i>	0.3945* (0.2028)	0.3171 (0.2342)
<i>FC</i>	−0.4583 (0.9104)	−0.2045 (1.0020)
<i>lnCapital</i>	−0.1501 (0.1765)	−0.0313 (0.1942)
<i>lnSale_cost</i>	−0.6121** (0.2638)	−0.5178* (0.2932)
<i>lnRD</i>	0.1658* (0.0939)	0.1546 (0.1133)
Firm fixed effects	Yes	Yes
Industry-year fixed effects	Yes	Yes
<i>N</i>	834	834
<i>R-square</i>	.9034	.8751

Note: Standard errors are in parentheses and clustered at a firm level.

*** $p < .01$; ** $p < .05$ and * $p < .1$.

innovation performance following the U.S. export control shock. The imposition of export control measures limits their access to advanced technology and products, compelling them to focus on independent innovation as a means to maintain their market share, ensure long-term development and overcome the crisis by achieving technological breakthroughs on their own. These

results indicate that the export control measures imposed by the United States have spurred a positive effect on the innovation performance of blacklisted Chinese firms. The limitations imposed by export control have seemingly incentivised these firms to intensify their efforts towards independent innovation, resulting in an observable improvement in their innovation capabilities.

4.2 | Robustness check

4.2.1 | Supplementing potential missing variables

Different cities have distinct business environments, and the level of policy support for firm innovation also varies across cities in China. To account for potential estimation bias resulting from differences in the cities where the firms are located, we additionally included city fixed effects in our regression model.

Table 6 presents a summary of the results regarding the impact of the entity list on the innovation performance of blacklisted Chinese firms, taking into account the control of city fixed

TABLE 6 Robustness check I.

	Control for the city fixed effect		PPML	
	(1)	(2)	(3)	(4)
	lnPatent	lnIPC	Patent	IPC
EntityList	0.4106** (0.1747)	0.4419** (0.1808)	0.7417*** (0.1117)	0.8093*** (0.1124)
lnAsset	1.5728 (1.1863)	1.1199 (1.3221)	6.5687*** (2.2041)	7.6154*** (2.1835)
lnAge	0.0573 (0.4643)	−0.1202 (0.5567)	−2.6776* (1.4723)	−3.0861** (1.5677)
TFP	0.2762 (0.1913)	0.1444 (0.2444)	0.1504 (0.4062)	0.0951 (0.4029)
FC	−0.4264 (0.9551)	−0.1410 (1.0502)	−4.8034*** (1.7778)	−5.7304*** (1.8156)
lnCapital	−0.0897 (0.1753)	0.0582 (0.1959)	−0.4589 (0.3067)	−0.4552 (0.2897)
lnSale_cost	−0.4827* (0.2590)	−0.3251 (0.3052)	0.2911 (0.4937)	0.4070 (0.5055)
lnRD	0.1497 (0.0958)	0.1302 (0.1151)	0.0218 (0.0567)	0.0465 (0.0605)
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	No	No
N	833	833	828	828
R-square	.9049	.8780	.9550	.9548

Note: Standard errors are in parentheses and clustered at the firm level.

***p < .01; **p < .05 and *p < .1.

effects. Columns 1 to 2 in Table 6 show that the estimated coefficients of the *Entitylist* variable remained positive and statistically significant across all specifications, reaffirming the robustness of our baseline findings.

4.2.2 | Re-estimation using alternative techniques

We also addressed the issue of zero-valued observations in our dependent variables by using the number of patents and IPC varieties before logarithm as the dependent variables. This approach, following the methods of Bhattacharya et al. (2022), helps mitigate estimation errors that can arise when estimating zero-valued observations using ordinary least squares (OLS). We also employed the Poisson pseudo-maximum-likelihood (PPML) estimation method in our study.

Columns 3 to 4 in Table 6 present the results using the PPML estimation method. Consistent with our previous findings, the coefficients of the variable '*Entitylist*' remained positive and statistically significant across all specifications. This indicates that our results hold even when using a different estimation method to account for zero-valued observations.

To address the potential issue with negative weights that may arise when employing a staggered adoption design in two-way fixed effects (FE) regressions, we adopted DID_M estimation, as outlined by Chaisemartin and Haultfoeuille (2020) and Mirenda et al. (2022). In addition, we also followed the approach of Beraja et al. (2023), where the weights of firms who have never experienced U.S. export control shocks in our sample were increased by a factor of 1000. The regression results are presented in Figures 4 and 5 and Table 7. The regression results continued significance underscores the robustness of our findings.

We also employed the two-stage difference-in-differences (2S-DID) regression method, following the approach outlined by Chaisemartin and Haultfoeuille (2020), to further examine the impact of U.S. export control policies on Chinese firms' innovation performance. Columns 1 to 2 of Table 8 display the results of the 2S-DID analysis. Consistent with our previous findings, the coefficients of the variable '*Entitylist*' remained positive and statistically significant across all specifications. This indicates that the U.S. export control policies have indeed exerted a positive influence on Chinese firms' innovation level.

The results of the 2S-DID analysis support our previous findings and further strengthen the evidence that the entity list and subsequent export control measures imposed by the U.S. have

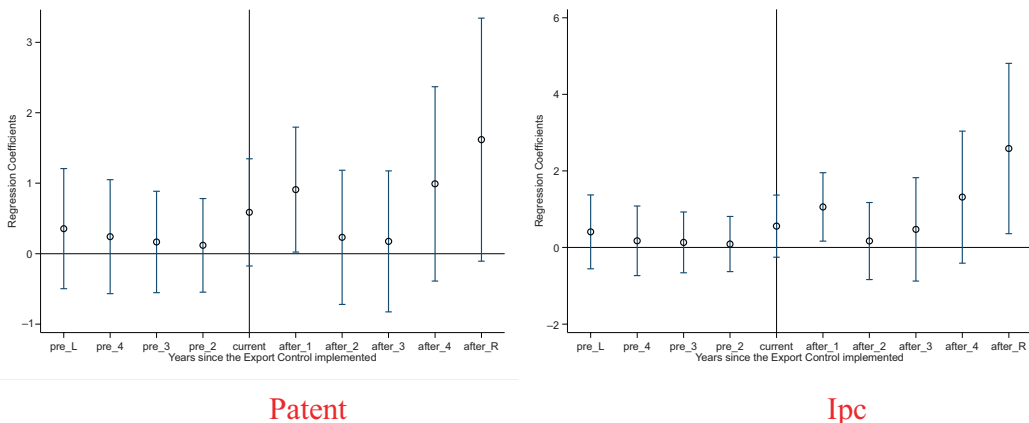


FIGURE 4 The Event Study results using the 'over-weighted' approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wjoe.13570)]

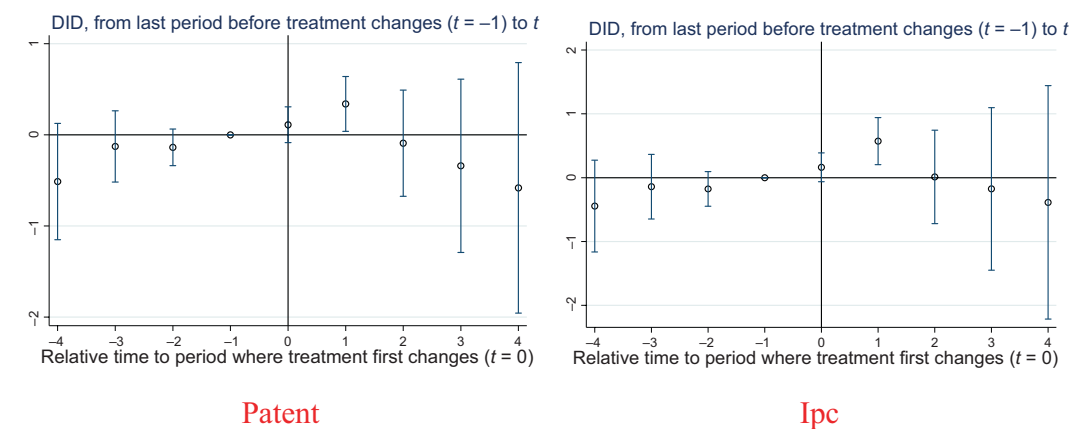


FIGURE 5 The Event Study results using the ‘DID_M’ approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

TABLE 7 Robustness check II.

	(1)	(2)	(3)	(4)	(5)
	Estimate	Standard Error	N	Controls	Fixed effects
Weight					
lnPatent	0.4787**	0.2379	834	Yes	Yes
lnIPC	0.5218**	0.2583	834	Yes	Yes
DID _M					
lnPatent	0.2077**	0.1030	344	Yes	Yes
lnIPC	0.2206**	0.1015	344	Yes	Yes

** $p < .05$.

prompted Chinese firms to enhance their innovation performance. These findings are robust and provide additional confidence in the positive impact of export control on Chinese firms’ innovation capabilities.

It is worth noting that the 2S-DID approach helps to address potential time-varying unobserved factors that could influence the outcomes. By comparing the changes in innovation performance between blacklisted and non-blacklisted firms before and after the implementation of export controls, the 2S-DID analysis provides a rigorous identification strategy to estimate the causal impact of the entity list on the innovation performance of the Chinese firms.

Overall, by controlling for city fixed effects and using alternative estimation methods, the robustness checks confirm the robustness of our findings, suggesting that the positive impact of the entity list on blacklisted Chinese firms’ innovation performance is robust and consistent.

4.2.3 | Reverse causality

To address potential endogeneity issues, particularly reverse causality, in our study, we employed a two-stage least squares (2SLS) regression with instrumental variables. The construction of the

TABLE 8 Further robustness checks.

	Two-stage DID		Instrumental variable		
	(1)	(2)	(3)	(4)	(5)
	lnPatent	lnIPC	EntityList	lnPatent	lnIPC
<i>EntityList</i>	0.4028** (0.2039)	0.5022*** (0.2060)		1.7739*** (0.4763)	1.8873*** (0.5325)
<i>IV_EntityList</i>			0.0986*** (0.0216)		
Kleibergen-Paap rk LM statistic			20.710***		
Kleibergen-Paap rk Wald F statistic			20.862		
10% maximal IV size			16.38		
Control variables	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	900	900	899	899	899
<i>R</i> -square	/	/	/	.5172	.5238

Note: Standard errors in parentheses and are clustered at the firm level.

*** $p < .01$; ** $p < .05$.

instrumental variable was based on the methodology proposed by Li et al. (2021) and comprised two components: (i) the inclusion of a firm in the CSI 300 Index and (ii) the voting divergence between China and the United States.

First, we adopted Aghion et al.'s (2013) approach, using whether a firm is included in the S&P 500 Index as an instrumental variable for researching firms' innovation. Analogously, the CSI 300 Index consists of the most representative 300 securities in the Chinese stock markets, launched on 8 April 2005, to reflect the overall performance of public firm securities in the Chinese stock markets. Adjustments to the securities sample occur approximately every 4–6 months. The inclusion of a public firm in the CSI 300 Index depends on its industry representation in the last 4–6 months, rather than on its business performance or investment potential, ensuring good exogeneity. Additionally, as the primary objective of U.S. export controls on China is to prevent the acquisition of advanced technology, public firms with industry representation may be more likely to attract such sanctions. Thus, this variable satisfies the relevance requirements for instrumental variables. We acquired all sample adjustment lists of the CSI 300 Index from China Securities Index Co., Ltd.,² and subsequently matched them with our sample to derive the variable $CSI300_{it}$. $CSI300_{it}$ takes the value of 1 if firm i is included in the sample of securities in the CSI 300 Index in year t ; otherwise, it is 0.

Second, the divergence in voting between China and the United States at the United Nations General Assembly was utilised as a metric to gauge Sino–U.S. relations. This metric, widely employed in numerous studies (Ambrocio & Hasan, 2021; Berger et al., 2013; Reynaud & Vauday, 2009), indicates greater political differences and a higher degree of deterioration in bilateral relations with higher divergence in voting. Poor bilateral relations increase the possibility of export controls. Simultaneously, the divergence in voting between China and the United States

²<https://www.csindex.com.cn/zh-CN/indices/index-detail/000300#/>.



does not have a specific impact on the operational behaviour of Chinese firms. To measure the divergence in voting, we used the ideal point distance data calculated by Bailey et al. (2017), Dreher et al. (2019) and Huang et al. (2023) to measure the degree of voting divergence between the two countries ($Vote_divergence_i$). A higher value of ideal point distance ($Vote_divergence_i$) indicates greater divergence between China and the United States, increasing the probability of the United States escalating the intensity of export controls on China. Consequently, our instrumental variable is defined as:

$$IV_EntityList_{it} = CSI300_{it} \times Vote_divergence_i \quad (6)$$

where the subscript i represents the firm and t represents the year. The higher the value of $IV_EntityList_{it}$, the greater the likelihood that firm i will be included in the U.S. entity list in year t .

Columns 3 to 5 in Table 8 outline the results of 2SLS estimation. The first-stage regression demonstrates a significant positive correlation between the instrumental variable and the U.S. export control policy. Subsequently, the second-stage results pass both the under-identification and weak-identification tests, confirming the reliability of the instrumental variable. The coefficients and their significance in the second stage further affirm the robustness of our baseline regression results.

These findings collectively underscore the effectiveness of the instrumental variable in mitigating potential endogeneity concerns. By employing 2SLS regression with instrumental variable, we successfully addressed the issue of reverse causality, obtaining robust results that support the positive impact of U.S. export control policies on the innovation performance of Chinese firms. The rigorous testing and validation procedures conducted in both stages enhance the credibility and reliability of our model's estimated coefficients.

4.2.4 | A placebo test

Performing a placebo test is a valuable approach for assessing the validity and robustness of our estimation. In this test, we created a random treatment group, which did not have any actual exposure to the export control policy, and estimated Equation (1) using this false treatment group. We repeated this process 500 times to generate a distribution of coefficients and t -values.

Figure 6 displays the kernel density estimation of the coefficients obtained from the placebo test, while Figure 7 shows the kernel density estimation of the corresponding t -values. These visualisations provide insights into the distribution of the estimated coefficients and t -values under the null hypothesis of no true treatment effect. If our estimation was valid, we would expect the coefficients and t -values to be centred around zero, indicating that there is no significant treatment effect in the false treatment group. However, if we observed a significant deviation from zero in the coefficients or t -values, our estimated treatment effect in the actual analysis might be biased or driven by other factors.

By conducting the placebo test and analysing the kernel density estimations in Figures 6 and 7, we were able to assess the credibility of our estimated treatment effect and determine whether it was statistically robust or potentially influenced by confounding factors.

The fact that the coefficients and t -values in the placebo test were centred around zero for the false treatment group, while the estimates for the true treatment group were at the end of the distribution on the right, provides strong evidence that the change in innovation performance

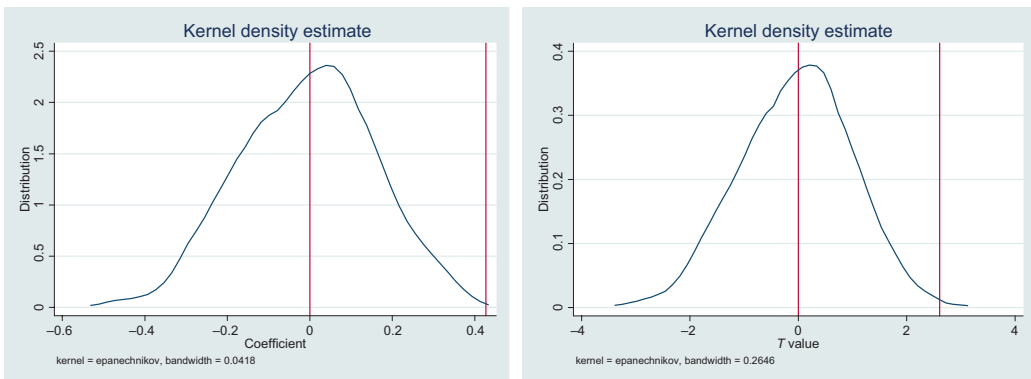


FIGURE 6 Kernel density estimation of the coefficient and t -value for $\ln\text{Patent}$. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

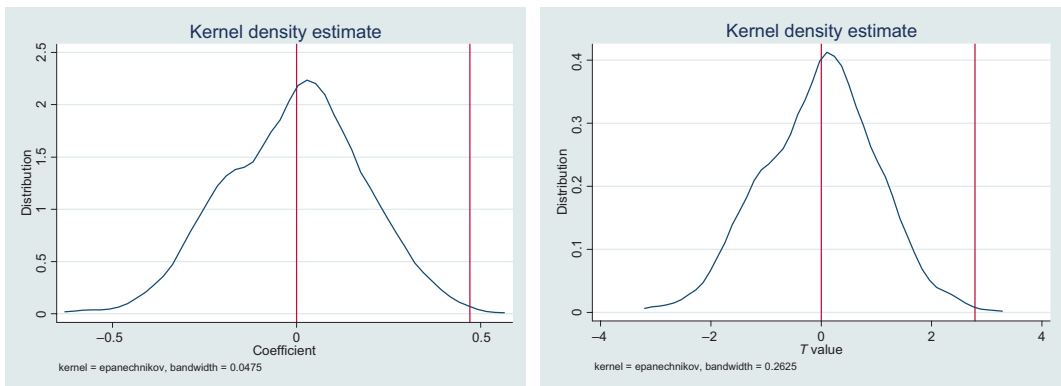


FIGURE 7 Kernel density estimation of the coefficient and t -value for $\ln\text{IPC}$. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

observed in the blacklisted Chinese firms can indeed be attributed to the impact of U.S. export control policies. The placebo test helps to establish the robustness of our baseline results by demonstrating that the estimated treatment effect was not driven by random variation or other unobserved factors. The fact that the true treatment group stands out from the placebo distribution further supports the conclusion that the observed effects on innovation performance were indeed caused by the implementation of U.S. export control measures.

Based on these findings, we can have increased confidence in the validity and reliability of our baseline results, as they demonstrate a consistent and significant impact of U.S. export control on the innovation performance of blacklisted Chinese firms.

4.3 | Mechanism analysis

Mechanism analysis is an important step in understanding the underlying factors that drive the impact of export controls on the innovation performance of blacklisted Chinese public firms. By identifying and exploring potential mechanisms, we can gain insights into the specific channels through which export control measures affect firms' innovation capabilities. Below, we separately consider the role of government grants, R&D investment and main business income.

4.3.1 | Government grants

In the context of ongoing international conflicts, a sanctioned country may opt to provide government grants to domestic firms as a means of addressing technical challenges arising from limited access to advanced technology. This strategic approach aims to foster the development and resilience of domestic industries. Government grants serve to mitigate the difficulties caused by disrupted supply chains, support firms in implementing necessary business adjustments and alleviate financial pressures faced by blacklisted firms. Additionally, external financial support can contribute to enhancing the quantity and quality of patents held by these firms (Acharya & Xu, 2017).

We utilised government grants (*Grants*) as the dependent variable in our model to examine whether export control measures promoted the allocation of government grants by the sanctioned country, thereby encouraging blacklisted firms to enhance their innovation performance. Data on government subsidies disclosed by 5289 Chinese public firms from 2003 to 2021 were obtained from CSMAR. The results, presented in column 1 of Table 9, demonstrate positive and significant coefficients for government grants. This finding indicates that export control measures increased the magnitude of government grants provided by the controlling country and

TABLE 9 Mechanism analysis.

	(1)	(2)	(3)
	<i>Grants</i>	<i>RDRatio</i>	<i>lnMain_sale</i>
<i>EntityList</i>	0.0908** (0.0454)	0.0401** (0.0195)	0.1156** (0.0525)
<i>lnAsset</i>	−1.4375*** (0.5320)	−0.1840 (0.1536)	1.1715*** (0.1504)
<i>lnAge</i>	−0.2596 (0.1763)	0.0507 (0.0866)	0.0696 (0.1539)
<i>lpacf</i>	−0.0403 (0.0403)	−0.2296** (0.1123)	1.0246*** (0.0264)
<i>SA</i>	1.2742*** (0.4442)	−0.1125* (0.0670)	−0.0511 (0.1150)
<i>lnCapital</i>	−0.0254 (0.0367)	0.0905** (0.0431)	−0.2890*** (0.0384)
<i>lnSale_cost</i>	0.0557 (0.0611)	0.1464 (0.1087)	0.0077 (0.0209)
<i>lnRD</i>	0.0051 (0.0235)	0.1368* (0.0779)	0.0091 (0.0090)
Firm fixed effects	YES	YES	YES
Industry-Year fixed effects	YES	YES	YES
<i>N</i>	821	834	834
<i>R-square</i>	.8430	.7322	.9970

Note: Standard errors are in parentheses and are clustered at the firm level.

****p* < .01; ***p* < .05 and **p* < .1.

that the innovation performance of the blacklisted firms greatly benefitted from these government grants.

4.3.2 | R&D investment

Traditionally, R&D investment is widely recognised as a crucial strategy for driving innovation (Trajtenberg, 1990). Increased investment in R&D enables firms to acquire advanced technology and develop new products and processes, leading to higher levels of innovation overall (Bilbao-Osorio & Rodríguez-Pose, 2004). Therefore, a key hypothesis regarding the impact of U.S. export control measures is that blacklisted Chinese public firms may increase their R&D investment in response to the heightened instability in the international supply chain. This strategy aims to alleviate the development bottleneck caused by external factors and prioritise independent innovation in technological advancement.

To assess the effect of export controls on R&D investment for innovation in blacklisted public firms, we calculated the proportion of R&D investment relative to sales revenue. This calculated ratio, referred to as R&D investment ratio (*RDRatio*), was utilised as a variable to gauge the intensity of R&D investment. Consequently, *RDRatio* served as the dependent variable in our baseline regression model. Column 2 in Table 9 presents the results. The positive and statistically significant coefficients on the R&D investment ratio provide compelling evidence that, following their blacklisting by the U.S. entity list, blacklisted firms exhibited a greater willingness to increase their R&D investment. This, in turn, enhanced their innovation performance compared to non-blacklisted Chinese firms.

4.3.3 | Main business income

The disruption of the supply chain inevitably leads to a certain degree of shock to firms' business status. Increased uncertainty in factors such as production and profitability creates an opportunity for blacklisted firms to streamline their operations and prioritise their core business activities. Through such strategic measures, these firms can enhance their core competitiveness, drive innovation and breakthroughs and stabilise their long-term development prospects.

To investigate the impact of U.S. export control sanctions on sanctioned firms' business operations, we utilised main business income data, representing revenue generated by firms from their primary production and operational activities. The logarithm of this variable, denoted as *lnMain_sale*, served as the dependent variable in our analysis. Main business income is a mandatory disclosure item in the annual reports of Chinese public firms, and the relevant data were obtained from CSMAR. The results, presented in Column 3 of Table 9, indicate that export controls prompted the sanctioned firms to shift their development focus towards their main business. This strategic shift proved to be beneficial in optimising resource allocation and fostering innovation.

Overall, the mechanism analysis shows that government grants enabled blacklisted firms to improve their innovation performance and contribute to the development and competitiveness of domestic industries. Moreover, the findings regarding R&D investment and main business income suggest that the U.S. export control measures compelled blacklisted Chinese firms to take strategic actions in response to limited access to advanced technology. These actions include increasing R&D investment and focusing on core business operations, which aim to enhance

innovation performance, reduce external dependencies and achieve long-term development and competitiveness despite export control-induced disruptions.

4.4 | Heterogeneity in the dependence of firms on imports from the United States

In this section, we examine whether the impact of export control measures on blacklisted firms varies depending on whether their production activities and operations rely on imports from the United States. The rationale is that the restrictions imposed by the U.S. entity list mainly affect the acquisition of U.S. technology or products through imports, while other forms of untraded technology licensing and cooperation may still be maintained in the short term. Therefore, it is expected that the impact of export controls on blacklisted firms will exhibit heterogeneity based on their reliance on imports from the United States.

We used the China Customs Import and Export database to identify the firms that relied on imports from the United States. Firms with records of imports from the United States in any period were considered to be firms that relied on imports from the United States throughout the entire sample period. By dividing the sample into firms that relied on imports from the United States and firms that did not, we examined whether there were differences in innovation performance among the blacklisted firms.

The results presented in Table 10 indicate that the estimated coefficients of *Entitylist* remained positive and statistically significant in the sample of firms that relied on imports from the United States. This suggests that blacklisted firms that depend on imports from the United States experience a notable increase in their innovation performance in response to export control measures. However, the coefficients were not statistically significant in the sample of firms that did not rely on imports from the United States. This implies that being included in the entity list does not significantly affect the innovation performance of firms that do not rely on imports of U.S. products or technologies.

TABLE 10 Heterogeneity in whether firms rely on imports from the United States.

	Firms rely on imports from the United States		Firms do not rely on imports from the United States	
	(1)	(2)	(3)	(4)
	lnPatent	lnIPC	lnPatent	lnIPC
<i>EntityList</i>	0.5226** (0.2205)	0.5957** (0.2231)	0.1531 (0.3204)	0.1287 (0.3371)
Control variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-Year fixed effects	Yes	Yes	Yes	Yes
<i>N</i>	532	532	211	211
<i>R</i> -square	.9038	.8770	.8843	.8503

Note: Standard errors are in parentheses and clustered at the firm level.

***p* < .05.

Overall, the findings support the hypothesis that export control measures primarily impact blacklisted firms that rely on imports from the United States, while firms that do not have such reliance do not exhibit significant differences in innovation performance when included in the entity list. This highlights the importance of considering the heterogeneity of firms' reliance on U.S. imports when analysing the effects of export control measures on innovation performance.

4.5 | Industrial diffusion effect

Due to the interconnectivity of industrial supply chains and the influence of peer effects, the innovation performance of firms operating in the same industry as blacklisted firms may also be affected (Foucault & Fresard, 2014; Leary & Roberts, 2014). Although the U.S. entity list included only 540 Chinese entities as of 2021, the resulting threats and uncertainties stemming from disruptions in the international supply chain due to export controls have had spillover effects on other firms in the same industries. To mitigate future uncertainties, these firms have started reducing their reliance on imported high-value inputs through independent innovation. Additionally, export control measures have prompted many firms in these industries to alter their business strategies and relocate their supply chains to domestic markets, reducing their dependence on foreign supply chains.

To further elucidate the impact of U.S. export controls on the innovation performance of Chinese firms, we expanded our sample to include all public firms within the same subindustries (excluding those already blacklisted by the entity list). This allowed us to analyse the industry-level impact of export control on innovation. As in the previous sections, we employed a multiple-period DID approach, treating firms in the same subindustries as blacklisted Chinese firms as the treatment group and public firms in other subindustries as the control group.

4.5.1 | Empirical results

First, we used the event study methodology to examine the presence of a parallel trend between the treatment and control groups prior to the implementation of U.S. export controls. The results of this analysis are shown in Figure 8.

The results reveal that there was no significant change in the trend of innovation performance before the implementation of the U.S. entity list. The coefficients approached zero in both cases, indicating that there was no substantial difference in innovation performance between the treatment and control groups prior to the policy shock. Therefore, the parallel trend assumption is satisfied. However, a significant difference in innovation performance emerged after the policy implementation, and this difference continued to strengthen over time.

To estimate the differences between the treatment and control groups before and after the implementation of the U.S. entity list, we employed a multiple-period DID approach. We defined '*ind_EntityList*' as an indicator variable representing whether firms in the same industry were subject to export controls in year t . The results are summarised in Table 11.

The findings demonstrate positive and significant coefficients for '*ind_EntityList*'. This suggests that after the implementation of U.S. export controls, the innovation performance of other firms in the same industries also improved. These results indicate that the impact of export controls on China extends beyond the firms blacklisted by the entity list and diffuses to the broader industry level.

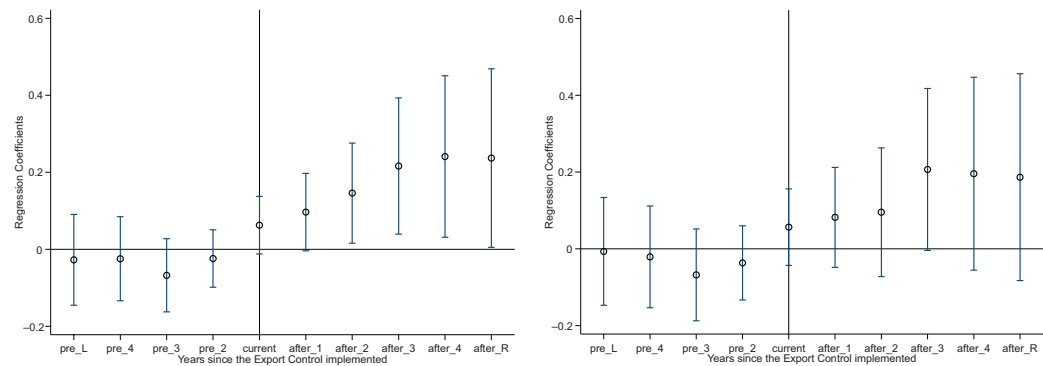


FIGURE 8 Parallel trend test of the firms in related industries directly affected by the U.S. entity list. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

TABLE 11 Multiple-period DID analysis of firms in the same industries as those blacklisted.

	(1)	(2)
	<i>lnPatent</i>	<i>lnIPC</i>
<i>ind_EntityList</i>	0.1134*** (0.0395)	0.0982** (0.0485)
<i>lnAsset</i>	0.7202*** (0.2463)	1.0037*** (0.2901)
<i>lnAge</i>	−0.0702 (0.1351)	−0.0301 (0.1610)
<i>TFP</i>	0.1704** (0.0678)	0.2153** (0.0848)
<i>FC</i>	−0.0210 (0.2101)	−0.1343 (0.2435)
<i>lnCapital</i>	−0.2030*** (0.0427)	−0.2486*** (0.0529)
<i>lnSale_cost</i>	−0.2167*** (0.0719)	−0.2956*** (0.0886)
<i>lnRD</i>	0.1626*** (0.0197)	0.1963*** (0.0243)
Firm fixed effects	Yes	Yes
Industry-year fixed effects	Yes	Yes
<i>N</i>	17,020	17,020
<i>R-square</i>	.7897	.7486

Note: Standard errors are in parentheses and clustered at the firm level.

*** $p < .01$; ** $p < .05$.

We conducted several robustness checks to validate the findings. First, we controlled for city fixed effects and employed Poisson pseudo-maximum-likelihood estimation and two-stage DID as alternative estimation methods. To tackle the challenges posed by negative weights in

staggered adoption design with two-way fixed effects, we also applied DID_M estimation methods and over-weighted approach. Additionally, we performed a placebo test to ensure the robustness of our results.

Tables 12 and 13 and Figures 9 and 10 present the results of these robustness checks, indicating that the coefficients of 'ind_Entitylist' remained positive and statistically significant. This confirms the robustness of our benchmark regression. Figures 11 and 12 display the results of the placebo test. Both the coefficients and *t*-values were centred around zero in all three scenarios. Furthermore, the estimated *t*-value of our true treatment group was located at the extreme right end of the distribution, indicating that our results are robust and not due to chance.

TABLE 12 Robustness check of firms in the same industry as blacklisted firms.

	Control for the city's fixed effect		PPML	
	(1)	(2)	(3)	(4)
	lnPatent	lnIPC	Patent	IPC
<i>ind_EntityList</i>	0.1144*** (0.0400)	0.0994** (0.0491)	0.1718*** (0.0583)	0.1778*** (0.0605)
<i>lnAsset</i>	0.7431*** (0.2498)	1.0201*** (0.2937)	1.5666*** (0.3814)	1.7763*** (0.4513)
<i>lnAge</i>	-0.0481 (0.1370)	-0.0074 (0.1631)	-0.4084 (0.2623)	-0.5472** (0.2746)
<i>TFP</i>	0.1619** (0.0690)	0.1968** (0.0865)	0.5367*** (0.1208)	0.4880*** (0.1197)
<i>FC</i>	-0.0537 (0.2137)	-0.1734 (0.2474)	-0.3962 (0.2819)	-0.6045* (0.3463)
<i>lnCapital</i>	-0.1896*** (0.0430)	-0.2282*** (0.0531)	-0.1816*** (0.0679)	-0.1854*** (0.0677)
<i>lnSale_cost</i>	-0.2124*** (0.0731)	-0.2838*** (0.0902)	-0.4144*** (0.1286)	-0.3757*** (0.1274)
<i>lnRD</i>	0.1660*** (0.0199)	0.2006*** (0.0246)	0.0324 (0.0296)	0.0287 (0.0322)
City fixed effects	Yes	Yes	No	No
	Two-stage difference-in-differences			
	lnPatent		lnIPC	
<i>ind_EntityList</i>	0.2180** (0.0843)		0.2402** (0.1030)	
Firm fixed effects	Yes		Yes	
Industry-year fixed effects	Yes		Yes	
<i>N</i>	17,016	17,016	16,705	16,705
<i>R-square</i>	.7913	.7504	.9267	.9249

Note: Standard errors are in parentheses and clustered at the firm level.

****p* < .01; ***p* < .05 and **p* < .1.

TABLE 13 Robustness check II.

	(1)	(2)	(3)	(4)	(5)
	Estimate	Standard error	N	Controls	Fixed effects
Weight					
lnPatent	0.1963***	0.0738	17,010	Yes	Yes
lnIPC	0.1455*	0.0879	17,010	Yes	Yes
DIDM					
lnPatent	0.0249***	0.0083	7405	Yes	Yes
lnIPC	0.0290***	0.0125	7405	Yes	Yes

*** $p < .01$ and * $p < .1$.

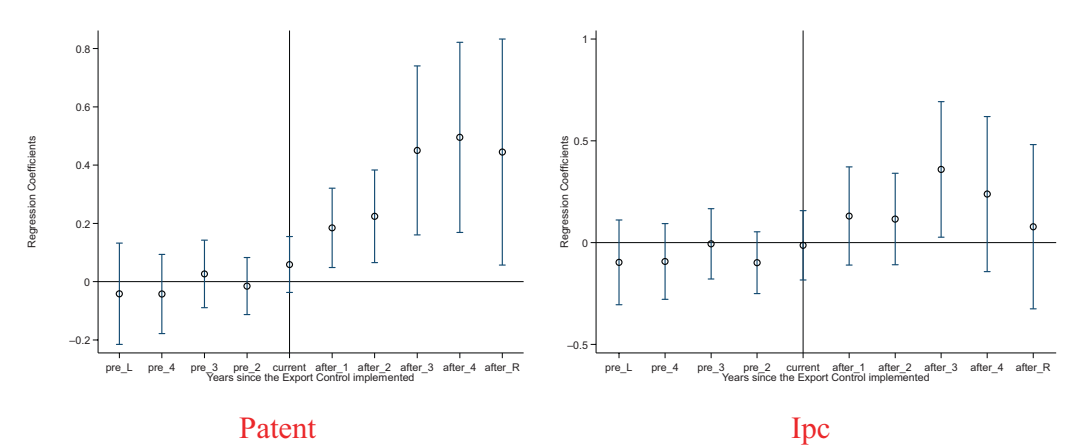


FIGURE 9 The Event Study results using the ‘over-weighted’ approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

4.5.2 | Mechanism analysis

In this section, we delve deeper into the mechanisms through which U.S. export control policies impact the innovation performance of Chinese public firms in related industries. Specifically, we examined the effects of export control on firms’ R&D investment ratio and government grants. The results presented in Table 14 reveal that the estimated coefficient of *ind_Entitylist* was positive and statistically significant. This indicates that U.S. export control policies stimulated R&D investment and government grants in related industries, ultimately leading to an enhancement in firms’ innovation performance.

4.5.3 | Heterogeneity in firms’ reliance on China’s domestic market

US export controls on Chinese firms have decreased the firms’ competitive edge in overseas markets, resulting in a shift in their market focus and supply chain towards China. This has intensified competition within the Chinese domestic market, prompting us to divide the sample based on firms’ sole reliance on China’s domestic market.

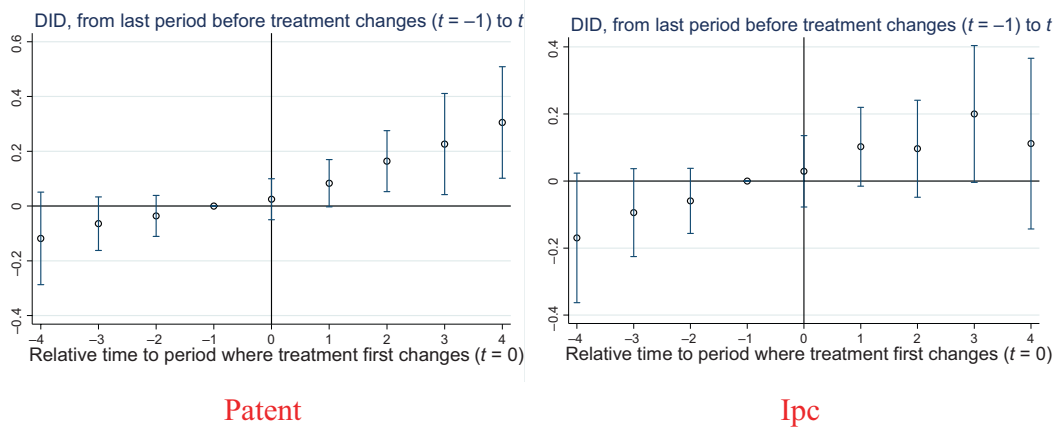


FIGURE 10 The Event Study results using the 'DIDM' approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

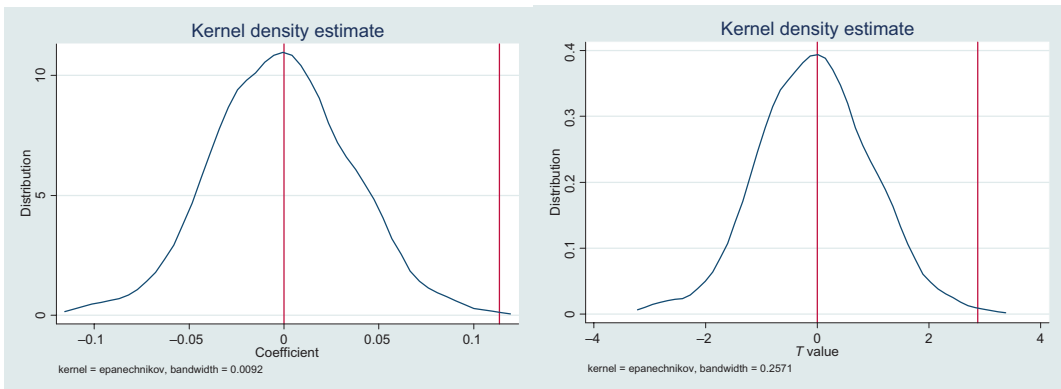


FIGURE 11 Kernel density estimation of the coefficient and t -value for $\ln Patent$. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

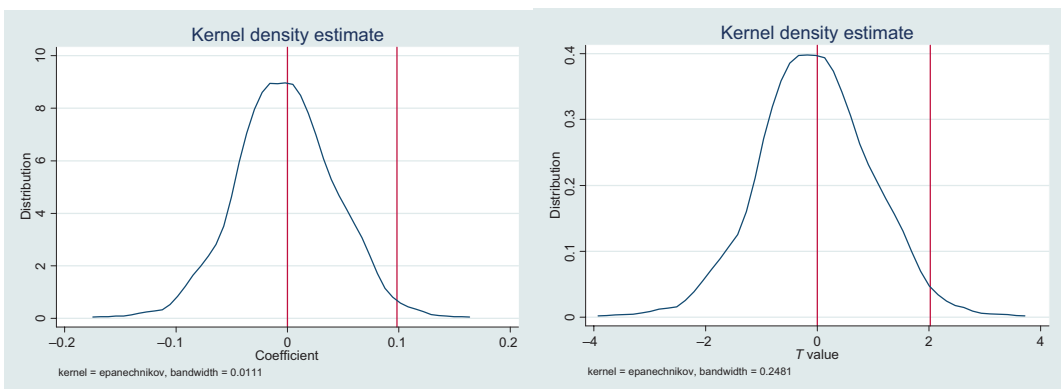


FIGURE 12 Kernel density estimation of the coefficient and t -value for $\ln IPC$. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

TABLE 14 Mechanism analysis of the effect on firms in same industry as those blacklisted.

	(1)	(2)
	<i>lnAmount</i>	<i>RDRatio</i>
<i>ind_EntityList</i>	0.0729** (0.0369)	1.5319*** (0.5564)
<i>lnAsset</i>	0.1296 (0.2631)	20.6944*** (6.5798)
<i>lnAge</i>	−0.0276 (0.1414)	3.1847 (4.7214)
<i>TFP</i>	−0.0155 (0.0575)	0.3379 (1.1777)
<i>FC</i>	0.4598** (0.2180)	−14.9368*** (5.5916)
<i>lnCapital</i>	−0.1412*** (0.0414)	−0.8088 (0.8467)
<i>lnSale_cost</i>	0.1260** (0.0623)	−3.8051*** (1.3578)
<i>lnRD</i>	0.1386*** (0.0228)	4.2945*** (0.7857)
Firm fixed effects	Yes	Yes
Industry-year fixed effects	Yes	Yes
<i>N</i>	16,830	11,274
<i>R-square</i>	.7665	.7634

Note: Standard errors are in parentheses and are clustered at the firm level.

*** $p < .01$; ** $p < .05$.

Table 15 presents the results of the heterogeneity analysis based on firms' reliance on China's domestic market. In the sample of firms that relied solely on China's domestic market, the estimated coefficients of *ind_Entitylist* remained positive and statistically significant. Conversely, these coefficients were not statistically significant in the sample of firms engaged in export or overseas business. This suggests that firms relying solely on China's domestic market face greater competition and exhibit stronger motivation to engage in innovation activities.

5 | THE REVERSE IMPACT OF THE ENTITY LIST ON U.S. SUPPLIERS

The influence of export control extends beyond Chinese firms that are blacklisted by the U.S. entity list; it also affects U.S. firms that have supply relationships with these blacklisted firms. Export control measures can lead to increased demand from Chinese firms for stockpiling key input products, aiming to mitigate potential risks and maintain short-term supply chain stability. However, strict regulations on export technology and the complex process of obtaining export licences have prompted Chinese firms to actively seek alternative suppliers and reduce their

TABLE 15 Heterogeneity in firms' reliance on China's domestic market.

	Firms have overseas business		Firms rely solely on domestic market	
	(1)	(2)	(3)	(4)
	<i>lnPatent</i>	<i>lnIPC</i>	<i>lnPatent</i>	<i>lnIPC</i>
<i>ind_EntityList</i>	0.0793 (0.0495)	0.0581 (0.0603)	0.1771*** (0.0668)	0.1646** (0.0829)
Control variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
<i>N</i>	10,692	10,692	6322	6322
<i>R</i> -square	.7956	.7580	.7843	.7367

Note: Standard errors are in parentheses and are clustered at the firm level.

*** $p < .01$; ** $p < .05$.

reliance on U.S. suppliers. As a result, U.S. suppliers have lost some Chinese customers, impacting their production and operational output and thereby inhibiting their innovation performance.

To further investigate the reciprocal impact of U.S. export controls on U.S. suppliers, we leveraged the FactSet database to identify U.S. suppliers of sanctioned Chinese entities. Using a multiple-period DID approach, we explored the effect of export controls on the innovation performance of these American suppliers.

5.1 | Empirical results

In this section, we utilised the event study methodology to conduct a parallel trend test (Figure 13). The results of the event study demonstrate that there was no significant change in the trend of innovation performance prior to the implementation of the U.S. entity list. The coefficient value approached zero during this period. However, a significant difference in innovation performance among U.S. suppliers emerged following the policy shock. Hence, we conclude that the parallel trend assumption is satisfied, affirming the appropriateness of our control group for this analysis.

We employed a multiple-period DID approach to estimate the differences between the treatment and control groups before and after the implementation of the U.S. entity list. In our analysis, we defined *US_Entitylist* as an indicator of whether the Chinese customers of U.S. public firms were blacklisted by the U.S. entity list in year t . The results of our analysis are presented in Table 16.

Our findings reveal that the estimated coefficients of *US_Entitylist* in relation to U.S. suppliers' innovation performance were negative and statistically significant. This suggests that the innovation performance of the U.S. public firms whose Chinese customers were affected by the entity list was significantly hindered. The decline in Chinese market share led to a reduction in demand for U.S. suppliers, subsequently impacting their operational performance.

5.2 | Robustness check

Consistent with earlier analyses, a robustness check was performed by introducing the state fixed effects, employing Poisson pseudo-maximum-likelihood estimation and utilising the two-stage

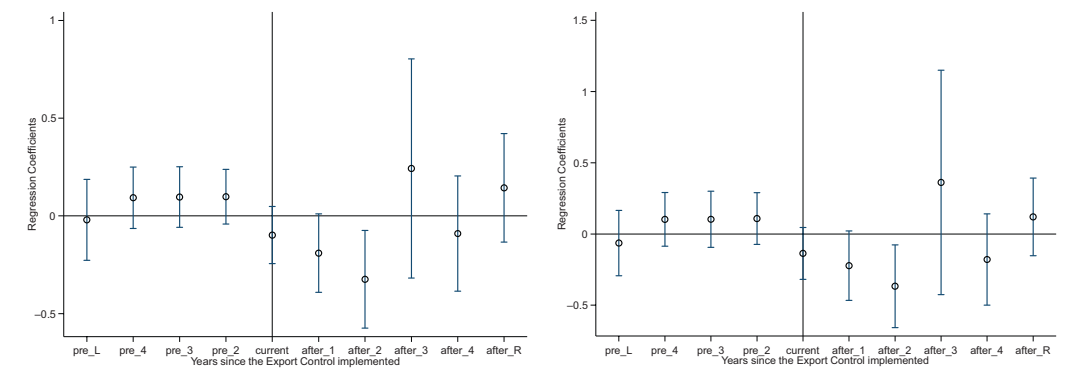


FIGURE 13 Parallel trend test of U.S. suppliers affected by the entity list. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 16 Multiple-period DID analysis of U.S. suppliers.

	(1)	(2)
	<i>lnPatent</i>	<i>lnIPC</i>
<i>US_Entitylist</i>	−0.2102** (0.1006)	−0.2298** (0.1139)
<i>lnAsset</i>	0.1644*** (0.0329)	0.2019*** (0.0383)
<i>TFP</i>	0.0002 (0.0199)	−0.0060 (0.0244)
<i>FC</i>	−0.0008 (0.0229)	−0.0086 (0.0255)
<i>lnCapital</i>	−0.1393*** (0.0353)	−0.1649*** (0.0414)
<i>lnSale_cost</i>	−0.0182 (0.0152)	−0.0205 (0.0183)
<i>lnRD</i>	0.0570*** (0.0168)	0.0733*** (0.0204)
Firm fixed effects	Yes	Yes
Industry-year fixed effects	Yes	Yes
<i>N</i>	8622	8622
<i>R-square</i>	.9001	.8924

Note: Standard errors are in parentheses and are clustered at the firm level.

*** $p < .01$, ** $p < .05$.

DID method. Furthermore, DID_M estimation and over-weighted approach were applied to address the issue of period heterogeneity in treatment effects and the number of treated units. To ensure the validity of our findings, a placebo test was conducted. The results, summarised in [Tables 17 and 18](#) and [Figures 14 and 15](#), indicate that the coefficients of *US_Entitylist* remained

TABLE 17 Robustness check of the effect on U.S. suppliers.

	Control for the state fixed effect		PPML	
	(1)	(2)	(3)	(4)
	<i>lnPatent</i>	<i>lnIPC</i>	<i>Patent</i>	<i>IPC</i>
<i>US_EntityList</i>	−0.2102** (0.1008)	−0.2298** (0.1142)	−0.4369** (0.1751)	−0.4321** (0.1773)
<i>lnAsset</i>	0.1644*** (0.0330)	0.2019*** (0.0384)	0.4768** (0.2172)	0.4527** (0.2056)
<i>TFP</i>	0.0002 (0.0200)	−0.0060 (0.0245)	−0.0548 (0.1625)	−0.0799 (0.1660)
<i>FC</i>	−0.0008 (0.0230)	−0.0086 (0.0256)	0.1079*** (0.0143)	0.1008*** (0.0157)
<i>lnCapital</i>	−0.1393*** (0.0354)	−0.1649*** (0.0415)	−0.4578** (0.1988)	−0.4094** (0.1968)
<i>lnSale_cost</i>	−0.0182 (0.0153)	−0.0205 (0.0183)	0.0384 (0.0902)	0.0817 (0.0842)
<i>lnRD</i>	0.0570*** (0.0169)	0.0733*** (0.0205)	0.2402* (0.1353)	0.2125* (0.1256)
State fixed effects	Yes	Yes	No	No
Two-stage difference-in-differences				
	<i>lnPatent</i>		<i>lnIPC</i>	
<i>US_EntityList</i>	−0.2369*** (0.0654)		−0.2464*** (0.0751)	
Firm fixed effects	Yes		Yes	
Industry-year fixed effects	Yes		Yes	
<i>N</i>	8622	8622	4545	4545
<i>R-square</i>	.9001	.8924	.9583	.9650

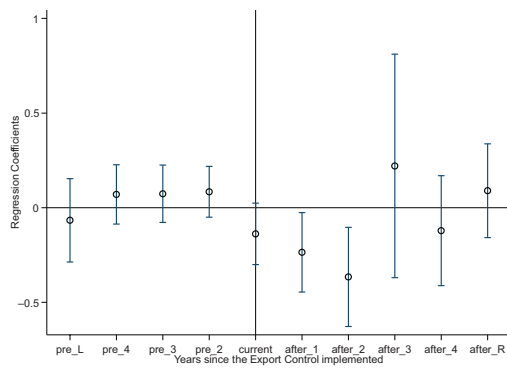
Note: Standard errors are in parentheses and are clustered at the firm level.

*** $p < .01$; ** $p < .05$ and * $p < .1$.

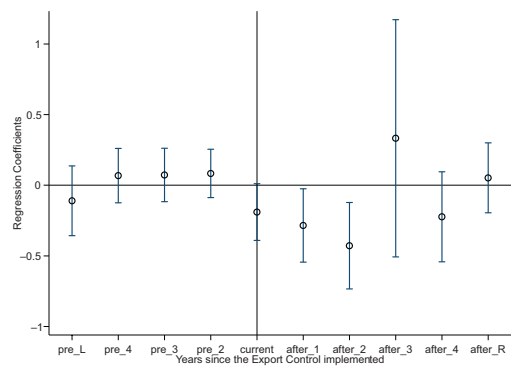
TABLE 18 Robustness check II.

	(1)	(2)	(3)	(4)	(5)
	Estimate	Standard error	<i>N</i>	Controls	Fixed effects
$\hat{\beta}_{FD}$					
<i>lnPatent</i>	−0.1629***	0.0572	7826	Yes	Yes
<i>lnIPC</i>	−0.1999***	0.0709	7826	Yes	Yes
<i>DID_M</i>					
<i>lnPatent</i>	−0.1033***	0.0343	2251	Yes	Yes
<i>lnIPC</i>	−0.1370***	0.0387	2251	Yes	Yes

*** $p < .01$.

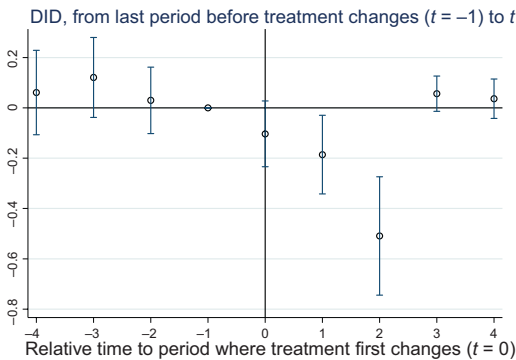


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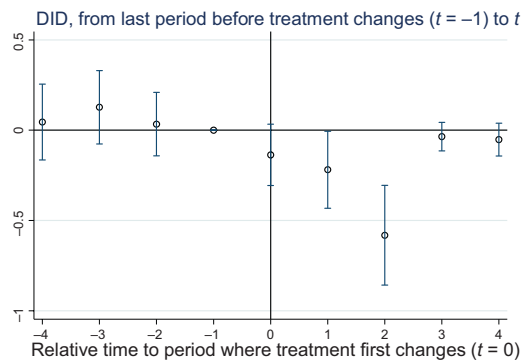


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FIGURE 14 The Event Study results using the ‘over-weighted’ approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]



Patent



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FIGURE 15 The Event Study results using the ‘DID_M’ approach. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/wee.13570)]

negative and statistically significant. This outcome enhances the robustness of our benchmark regression in this section.

To assess the robustness of our baseline results, we created a random treatment group and applied our baseline model using this false treatment group. This process was repeated 500 times to mitigate the potential influence of individual factors specific to our actual treatment group. The kernel density estimation of the coefficients and *t*-values obtained from these iterations are presented in Figures 16 and 17 respectively. These figures demonstrate the consistency and robustness of our baseline findings.

5.3 | Mechanism analysis

In this section, we examine the potential mechanisms through which U.S. export controls may hinder the innovation performance of U.S. suppliers. Previous research has highlighted the significance of firms' profitability and return on capital employed (*ROCE*) in driving their

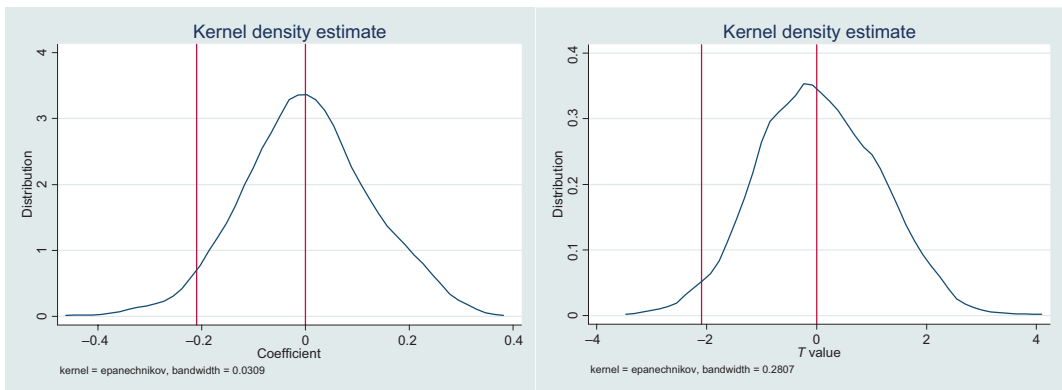


FIGURE 16 Kernel density estimation of the coefficients and t -values for $\ln Patent$. [Colour figure can be viewed at wileyonlinelibrary.com]

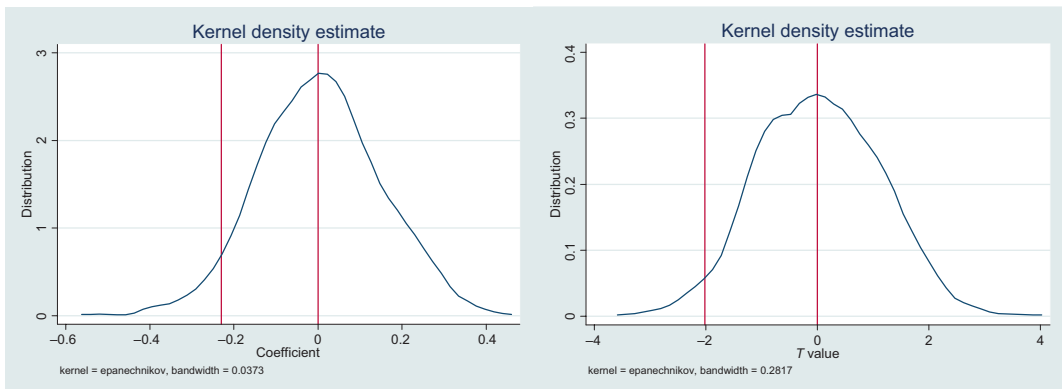


FIGURE 17 Kernel density estimation of the coefficients and t -values for $\ln IPC$. [Colour figure can be viewed at wileyonlinelibrary.com]

innovation levels (Asni & Agustia, 2022; Audretsch, 1995; Battisti et al., 2010). Higher levels of ROCE and profitability indicate increased value creation, which forms a crucial foundation for establishing firms' competitive advantage and facilitating investment in innovation (Liargovas & Skandalis, 2010). Consequently, we employed ROCE and *Net_Income* as variables to measure the return on capital employed and profitability of U.S. suppliers respectively.

Table 19 presents the summarised results. The estimated coefficients of '*US_Entitylist*' in both columns were negative and statistically significant. This indicates that the implementation of export controls has resulted in a decline in market share for U.S. suppliers in China, impacting their ROCE, profit and innovation performance.

6 | CONCLUSION AND POLICY IMPLICATIONS

This paper investigated the impact of U.S. export controls on the innovation performance of Chinese and related U.S. firms. We employed propensity score matching techniques and a multiple-period difference-in-differences approach to analyse the data.

**TABLE 19** Mechanism analysis of the effect on U.S. suppliers.

	(1)	(2)
	<i>Net_Income</i>	<i>ROCE</i>
<i>US_EntityList</i>	−0.3330** (0.1664)	−0.6737** (0.3381)
<i>lnAsset</i>	0.2056** (0.0995)	0.6266 (0.6502)
<i>TFP</i>	0.1230*** (0.0375)	0.4617* (0.2384)
<i>FC</i>	−0.2420 (0.2901)	−0.0086 (0.0305)
<i>lnCapital</i>	−0.2065 (0.1415)	0.0939 (0.3524)
<i>lnSale_cost</i>	−0.0038 (0.0301)	−0.2871 (0.4192)
<i>lnRD</i>	0.0777 (0.0728)	0.3049 (0.3627)
Firm fixed effects	Yes	Yes
Industry-year fixed effects	Yes	Yes
<i>N</i>	8622	7010
<i>R-square</i>	.7733	.1836

Note: Standard errors are in parentheses and are clustered at the firm level.

*** $p < .01$; ** $p < .05$ and * $p < .1$.

Our analysis yields three main conclusions. First, the U.S. entity list has prompted blacklisted Chinese firms to improve their innovation performance, particularly for firms reliant on imports from the United States. We found that the U.S. entity list influences Chinese government grants, increases R&D investment in blacklisted firms and encourages these firms to refocus their development efforts on their core business. Second, the implementation of the U.S. entity list also positively affects the innovation performance of Chinese public firms operating in the same industries as blacklisted firms. It leads to increased allocation of Chinese government grants and higher R&D investment, especially for firms that solely rely on China's domestic market. Third, the U.S. export control measures have a negative impact on the innovation performance of U.S. suppliers. These suppliers experience reduced income, profit and return on employed capital. Our robust analysis, accounting for city fixed effects and employing appropriate estimation methods, instrumental variables and placebo tests, supports this finding.

Our findings have important practical and policy implications. It is crucial for sanctioned countries to diversify their trade relationships and reduce their dependence on countries that implement export controls. Governments can support firms by offering subsidies to mitigate potential negative impacts and encourage increased investment in key technologies and products.

For countries implementing export controls, it is essential to carefully assess the policy's effects and consider the potential negative impact on innovation performance. Unilateral implementation of such policies may have unintended consequences. Our findings provide evidence of the costs associated with unilateralism and highlight the motivation for sanctioned countries to overcome challenges and achieve technological breakthroughs.

ACKNOWLEDGEMENTS

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from China Stock Market & Accounting Research Database (CSMAR). Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the authors with the permission of CSMAR.

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